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## 3.4 Audiology

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### 3.4.1 Basic Audiologic Assessments

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#### ◆ Key Features

- Pure tone testing—air and bone conduction thresholds
- Speech testing—speech reception thresholds (SRT) and word recognition ability
- Immittance battery—tympanometry, acoustic reflex thresholds, acoustic reflex decay, auditory tube function assessment

Air and bone conduction pure tone testing determines the type, degree, and configuration of hearing loss. Thresholds represent the softest sound audible at least 50% of the time, measured in decibels (dB). Speech reception thresholds (SRT) serve to verify pure tone thresholds. Word recognition ability represents an individual's ability to discriminate speech. Immittance testing assesses the status of the tympanic membrane, middle ear, and acoustic reflex pathways.

#### ◆ Epidemiology

Approximately 28 million Americans have hearing loss. Approximately 314 in 1,000 people over age 65 have hearing loss. Forty to fifty percent of people over the age of 75 suffer from hearing loss.

#### ◆ Clinical

##### Signs and Symptoms

Persons with hearing loss frequently report difficulty hearing in background noise. They mistakenly feel others around them are mumbling. They often need the television volume louder than is comfortable for others and ask for things to be repeated. Many experience tinnitus.

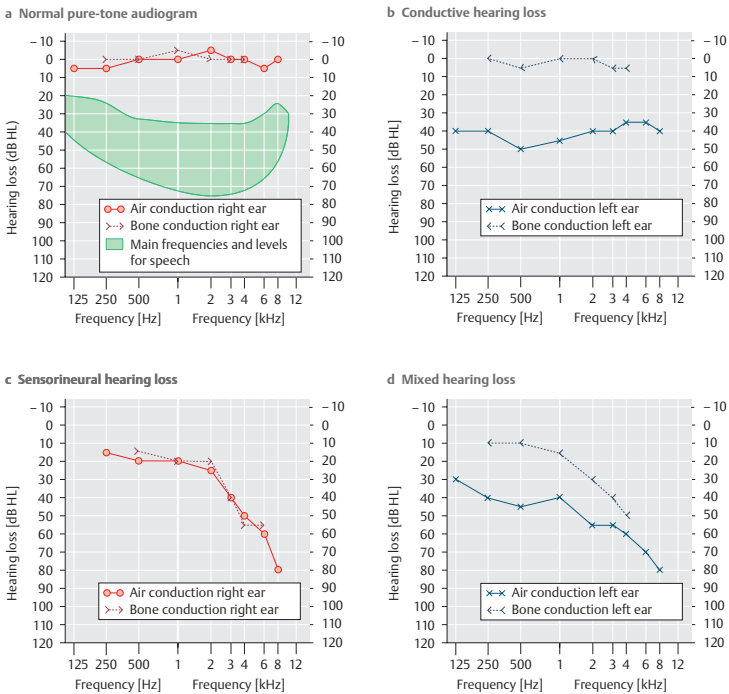
##### Differential Diagnosis

Results of audiometric testing are typically plotted on an audiogram (**Fig. 3.7**). An audiogram is a graph with frequency (in Hz) plotted on the x axis and intensity (in dB) plotted on the y axis. It is necessary to employ masking (a static type of noise) in certain situations to prevent the nontest ear from detecting the test signal.

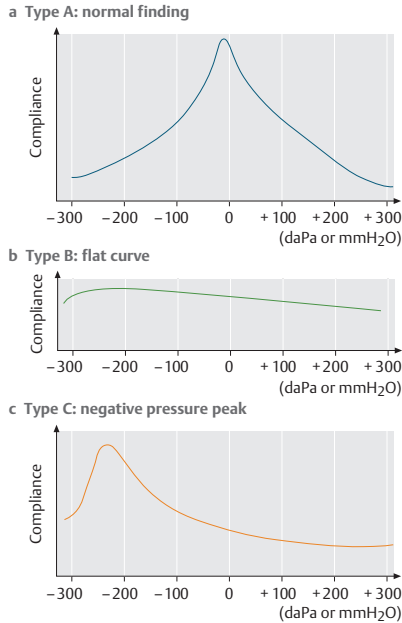
Results of the audiogram and tympanogram (**Fig. 3.8**) may suggest various forms of otologic disease. Red flags should prompt referral to an otolaryngologist (discussed below). Otologic disease may be categorized as conductive hearing loss, sensorineural loss (SNHL), or mixed hearing

loss. In a conductive loss, there is an air–bone gap on the audiogram, with better bone conduction thresholds, with appropriate masking. The loss may be due to dysfunction involving the external canal, tympanic membrane, or ossicular chain or to middle ear fluid. In a sensorineural loss, bone conduction thresholds typically equal air conduction, and the loss may be due to dysfunction at the level of the cochlea, CN VIII, or central pathways. A poor word-recognition score, especially out of proportion to pure tone thresholds, suggests possible retrocochlear pathology, such as an acoustic tumor.

Patterns of SNHL may be suggestive of certain etiologies, taken together with clinical history and exam. For example, acoustic trauma (noise-induced loss) often results in a loss centered around 4 kHz with recovery at higher frequency (i.e., a 4-kHz notch); presbycusis often involves symmetric downsloping loss, worse at higher frequency; a fluctuating low-frequency loss is often seen with Ménière's disease.



**Fig. 3.7** Pure-tone threshold audiometry. **(a)** Normal pure-tone audiogram. **(b)** Audiogram showing conductive hearing loss. **(c)** Audiogram showing sensorineural hearing loss. **(d)** Audiogram showing mixed hearing loss. (Used with permission from Probst R, Grevers G, Iro H. *Basic Otorhinolaryngology: A Step-by-Step Learning Guide*. Stuttgart/New York: Thieme; 2006:179.)



**Fig. 3.8** Normal and abnormal tympanogram patterns. **(a)** The normal tympanogram has a prominent, sharp peak between +100 and -100 daPa. **(b)** The type B tympanogram is flat or has a very low, rounded peak. This indicates immobility of the tympanic membrane, which may be due to fluid in the middle ear or tympanic atelectasis. A type B tympanogram with large volume is seen with the perforation. **(c)** The type C tympanogram has a peak in the negative pressure region below -100 daPa, consistent with impaired middle ear ventilation. (From Probst R, Grevers G, Iro H. *Basic Otorhinolaryngology: A Step-by-Step Learning Guide*. Stuttgart/New York: Thieme; 2006:185.)

The Jerger system is commonly used to classify tympanometric results:

- Type A: Normal
  - Pressure peak +100 to -100 daPa
  - Immittance 0.3 to 1.7 mL
- Type As: Shallow
  - Pressure peak +100 to -100 daPa
  - Immittance < 0.3 mL
- Type Ad: Deep
  - Pressure peak +100 to -100 daPa
  - Immittance > 1.7 mL
- Type C: Negative pressure
  - Pressure peak ≤ 100 daPa
  - Immittance normal or shallow

- Type B: Flat
  - No pressure peak or immittance
  - Small ear canal volume suggests ear canal occlusion.
  - Normal volume suggests otitis media.
  - Large ear canal volume suggests patent pressure equalization tube or perforated eardrum.

## ◆ Evaluation

### Tuning Fork Tests

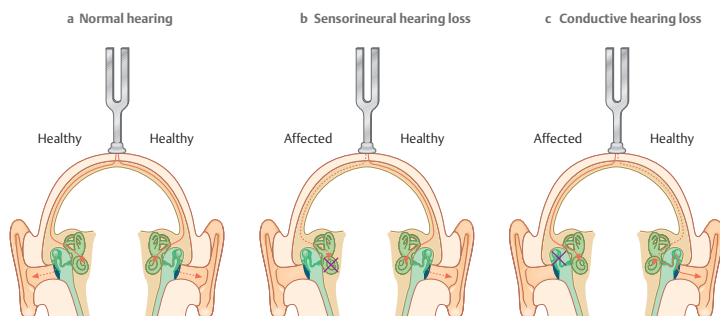
The goal of these tests, including the Weber test (**Fig. 3.9**), is to differentiate between conductive and SNHL. The tuning forks vibrate at a fixed frequency between 256 and 1024 Hz. To test bone conduction, the fork is placed firmly on the skull bone. Typically, a 512-Hz tuning fork is used.

### Weber Test

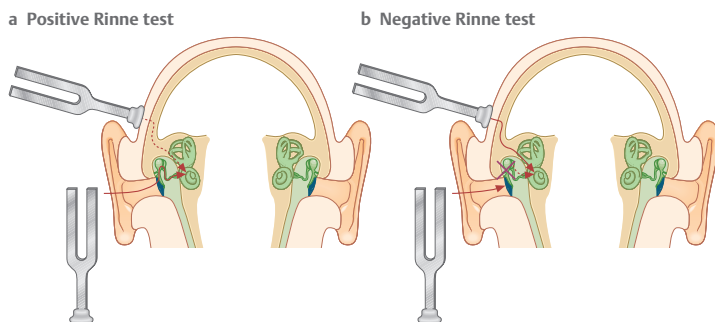
The Weber test will lateralize to the ear with worse conductive loss if sensory hearing is equal in both ears (**Fig. 3.9**). Alternatively, the Weber test will lateralize to the ear with better sensory hearing if there is no conductive hearing loss in either ear.

### Rinne Test

The Rinne test (**Fig. 3.10**) compares the level of air conduction and bone conduction in the same ear. In a normal ear, air conduction is greater. It usually takes about a 30-dB conductive loss to flip the tuning fork response to bone conduction greater than air.



**Fig. 3.9** The Weber test. A vibrating tuning fork is placed on the midline of the skull. **(a)** When hearing is normal, the sound is perceived with equal loudness in or between both ears. **(b)** With unilateral sensorineural hearing loss, the sound is lateralized to the better ear. **(c)** With unilateral conductive hearing loss, the sound is lateralized to the affected side. (Used with permission from Probst R, Grevers G, Iro H. *Basic Otorhinolaryngology: A Step-by-Step Learning Guide*. Stuttgart/ New York: Thieme; 2006:168.)



**Fig. 3.10** The Rinne test. Air and bone conduction are compared in the same ear to determine the auditory threshold for the tuning fork and/or its loudness. **(a)** In the absence of conductive hearing loss, air conduction is perceived as being louder and/or of longer duration than bone conduction. **(b)** When conductive hearing loss is present, bone conduction is perceived as being louder and/or more prolonged than air conduction. (Used with permission from Probst R, Grevers G, Iro H. *Basic Otorhinolaryngology: A Step-by-Step Learning Guide*. Stuttgart/New York: Thieme; 2006:69.)

## Physical Exam

Test results are best obtained in a sound-treated booth. Insert earphones placed in the ear canal are the transducer of choice in assessing air conduction thresholds due to infection control, increased interaural attenuation, and prevention of collapsed ear canals. Supra- and circumaural headphones are options as well in certain situations. Bone conduction is assessed by placing an oscillator on a mastoid bone. Forehead placement is another option in special situations.

Conventional behavioral audiometry is conducted by having a person respond to pure tones (e.g., raising a hand, pushing a button). SRTs are obtained by repeating spondee (two-syllable words with equal weight on both syllables; e.g., hotdog, baseball). Word recognition is assessed by repeating phonetically balanced lists of monosyllabic words at suprathreshold intensity (approximately 30–40 dB above the SRT).

In tympanometry, the maximum compliance peak is determined by presenting a probe tone (typically 1,000 Hz in infants up through 6 months of age, 226 Hz in all others) in the sealed ear canal while varying air pressure (+200 to –400 daPa). Maximum compliance occurs when the middle and external ear air pressures are equal.

## Other Tests

Acoustic reflex thresholds and acoustic reflex decay can be measured both ipsilaterally and contralaterally and across different frequencies. A probe tone is presented in either ear while a change in immittance is monitored.

This assesses the acoustic pathways by measuring the contraction of the stapedius muscle in response to acoustic stimulation. Auditory (eustachian) tube function testing reveals how well middle ear pressure is equalized. Testing occurs in the same manner as basic tympanometric testing, but with

a person pinching the nose and swallowing in between artificial changes in ear canal air pressure (Toynbee test).

## Pathology

Hearing loss may be congenital, acquired (e.g., exogenous sources such as perinatal infection) or genetic (hereditary). Genetic hearing loss may be syndromic or nonsyndromic. Acquired hearing loss is frequently related to the aging of the auditory system (presbycusis). It can also occur from trauma, infection, excessive noise exposure, and ototoxic medications. Hearing loss may remain stable over time, sudden, progressive, or fluctuating.

## ◆ Treatment Options

Patients with unilateral or conductive hearing losses should be referred for medical evaluation. Other red flags should prompt otolaryngology referral, such as evidence of a sudden hearing loss, rapidly progressing or fluctuating sensorineural loss, word recognition score poorer than expected based on pure tone thresholds, word recognition score asymmetry greater than 10%, evidence of middle ear dysfunction, evidence of otorrhea, or any hearing loss in a child.

## ◆ Outcome and Follow-Up

Following medical clearance, a variety of amplification devices are available and appropriate for most persons with hearing loss.

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## 3.4.2 Pediatric Audiologic Assessments

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### ◆ Key Features

- The goal of universal newborn hearing screening is to have all infants screened by 1 month, comprehensive audiologic evaluation when needed by 3 months, and intervention in infants with confirmed hearing loss by 6 months.
- There are more stringent cutoffs for normal hearing thresholds in pediatric patients: 15- to 20-decibel (dB) hearing loss, as opposed to 25-dB hearing loss in adults.
- Children require specialized testing that is administered by a well-trained pediatric audiologist. It is essential to use the cross-check principle when evaluating children in order to ensure a proper diagnosis.
- Testing protocols often need to be modified to obtain the same audiologic information readily acquired from cooperative adults.

Prior to the implementation of newborn hearing screenings, children with hearing loss were often misdiagnosed or identified later in life. When the diagnosis and treatment are delayed, the impact of hearing loss in children can

result in speech, language, cognitive, social, and academic delays. Early Hearing Detection and Intervention (EHDI) programs have become the standard of care in the United States, and screening for hearing loss occurs in 95% of infants born in the United States. Screening of neonates for hearing loss should be performed with automated distortion product otoacoustic emissions (DPOAE) or auditory brainstem response testing (ABR) prior to discharge from hospital.

The goal of universal newborn hearing screening is to have all infants screened by 1 month of age, comprehensive audiologic evaluation when needed before 3 months of age, and early intervention in infants with confirmed hearing loss by 6 months of age.

### ◆ Epidemiology

The prevalence of congenital hearing loss in newborns is about two to three infants per 1,000, or approximately 13,000 babies born in the United States each year with some degree of permanent hearing loss. Most of these babies are otherwise healthy and have no family history of hearing loss.

### ◆ Clinical

Even mild, untreated hearing loss in infants and young children can significantly impact their speech, language, social, cognitive, and academic development. As children develop, optimal learning and communication require hearing the speaker clearly, developmentally appropriate language, and speech levels that are at least 20 decibels (dB) above the background noise. Children with unidentified hearing loss will have considerable challenges, and as mentioned, these can greatly hinder a child from reaching his or her potential. Any concerns expressed by caregivers should be formally evaluated. Informal assessments are not adequate in evaluating a child's hearing.

A thorough birth history should be ascertained from parents, including history of prematurity, prenatal/postnatal infections, hyperbilirubinemia, exposure to ototoxic antibiotics, hypoxia, and encephalopathy. A developmental history is important to distinguish an isolated hearing loss from a more systemic problem causing speech delay. Syndromic causes and a family history of hearing loss should also be determined. A history of ear infections may indicate middle ear disease and conductive hearing loss. Further information can be found in **Chapter 8.11**.

### ◆ Evaluation

#### Physical Exam

A full head and neck physical exam concentrating on the otologic exam should be performed. When possible, an otomicroscope should be used to remove any cerumen or debris from the ear canal. The external auditory canal, tympanic membrane, and associated landmarks are assessed. Pneumatic otoscopy should be performed to determine mobility of the tympanic membrane and presence of tympanic membrane perforation or middle ear fluid. Craniofacial abnormalities may indicate a syndromic cause of hearing loss.

## Testing

Depending on the developmental age of the child, behavioral observation, visual reinforcement audiometry, conditioned play audiometry, and conventional audiometry are utilized to obtain pure tone thresholds. Behavioral observation testing is performed in infants up to 6 months of age. Visual reinforcement audiometry is done in children between 6 months and ~3 years of age. Conditioned play audiometry can be performed in children between the ages of 3 and 6. Last, conventional audiometry is performed in children that are older than 6 years of age.

With any of these tests, results are most reliable when testing is conducted with two people (i.e., one person performing the test protocol and one person in the booth to redirect the child—it is typically not advantageous to use a child's parent for this role). Testing with insert earphones is preferred so that ear-specific information can be obtained (headphones may also be used in some instances, but caution is needed to avoid collapsing young ear canals). Testing can also be conducted in the sound field; however, ear-specific information will not be known. Warbled tones and narrowband noise are frequently used in testing children in addition to pure tones.

In behavioral observation testing, examiners look for any physical changes in a child in response to sound (i.e., blinking or other eye movement, cessation of sucking). In visual reinforcement audiometry, children are rewarded (e.g., a lighted box with an animated character) by turning their head in response to sound. In play audiometry, children are conditioned to perform a task (e.g., drop blocks in a bucket, put pegs on a board) in response to pure tones. Conventional audiometry requires the patient to raise a hand or press a button when a sound is heard.

Speech awareness thresholds may be obtained in place of speech reception thresholds. Picture boards or body part pointing can also help in obtaining speech information. High-frequency probe tone (1,000 Hz) testing is needed in tympanometry to appropriately assess the middle ear function in children who have a developmental age of 6 months or younger. In some children, behavioral information should be confirmed through objective/electrophysiological testing. If the child has severe cognitive impairment or is otherwise unable or too young to cooperate for behavioral testing, objective testing should be obtained via otoacoustic emissions (OAE) and auditory brainstem response (ABR) evaluations.

OAE is an objective test to determine functional capacity of the outer hair cells of the cochlea. When sound is delivered to the cochlea, it produces a low-intensity sound called otoacoustic emissions. There are several types of otoacoustic emissions, however. Distortion product OAE (DPOAE) testing is widely used due to the ability to test at different frequencies. A probe is placed in the ear canal and two simultaneous pure tones at two different frequencies and intensities are delivered to the ear. Emissions are recorded at different frequency levels. DPOAEs should be interpreted with caution, as results are affected by obstructed external auditory canal, positioning and seal of the probe, tympanic membrane perforations, and middle ear disease.

ABR testing is an objective test that evaluates the function of the auditory nerve and brainstem elicited after an auditory stimulus. The interpeak latency (time between peaks) and interaural latency (difference between



ears) are also assessed. Each waveform in ABR correlates to a structure on the auditory pathway. In general, waves correspond to the eighth cranial nerve, cochlear nucleus, olivary complex (superior), lateral lemniscus, and inferior colliculus (mnemonic: ECOLI). Different frequencies and intensities are tested until a reproducible waveform is identified. Further information regarding DPOAEs and ABRs is found in **Chapter 3.4.3**.

### ◆ Treatment Options

The priority in children is to obtain appropriate auditory rehabilitation as soon as the problem is identified. Because plasticity within the auditory cortex is finite and diminishes dramatically after the first few years, a failure to obtain auditory input within this critical period will result in cortical deafness: an inability of the brain to process auditory input adequately later in life if aided or implanted. Thus, if hearing loss cannot be corrected by treating obvious causes, such as a myringotomy tube for middle ear fluid, amplification should be obtained. If there is minimal or no benefit from appropriate amplification, cochlear implant (CI) evaluation should be sought as soon as possible.

There are many modifications available to assist in the fitting of hearing aids to even very young infants and children. To accommodate ear growth, children are fitted with behind-the-ear hearing aids, and their earmolds need to be replaced frequently. In addition to hearing aids, children may benefit from the use of hearing assistive technology (e.g., FM system) in classrooms to improve the signal-to-noise ratio. Hearing assistive technology provides a handheld microphone for teachers to talk into, which transmits the sound of the teacher's voice directly to child's hearing aid.

### ◆ Outcome and Follow-Up

Children diagnosed with hearing loss should receive an otologic consult, an ophthalmologic consult, possibly genetic counseling, and routine monitoring of their hearing abilities. Depending on the origin of the hearing loss, siblings of diagnosed children should also have their hearing assessed. Children with high risk factors for hearing loss should receive periodic reevaluations as well. Infants who pass the newborn hearing screening but have risk factors for hearing loss should have at least one diagnostic audiology assessment by 24 to 30 months of age.

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## 3.5 Hearing Loss

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### 3.5.1 Conductive Hearing Loss

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#### ◆ Key Features

- Conductive hearing loss (CHL) occurs when sound is not efficiently transmitted through the external auditory canal, tympanic membrane, and ossicles to a functioning cochlea.
- CHL can occur due to dysfunction of the external auditory canal (EAC), the tympanic membrane, or the ossicles.
- Most cases of CHL can be surgically corrected.

Disorders of the external auditory canal (EAC), tympanic membrane (TM), middle ear, and ossicles can impair sound transmission to a functioning cochlea, resulting in conductive hearing loss (CHL).

#### ◆ Clinical

##### Signs

Various forms of obstruction can occur at the level of the EAC (e.g., cerumen, foreign body, congenital canal atresia, inflammatory stenosis with acquired atresia, bony exostosis, canal cholesteatoma). Blockage of the EAC can cause up to a 40-dB hearing loss. The TM may be thickened from scarring or tympanosclerosis; it can be retracted from negative middle ear pressure and drape on the incus and stapes (e.g., myringo-incudostapediopexy) or even onto the cochlear promontory. In addition, perforations of any size and middle ear effusion may interfere with the proper vibration of the ear drum and cause hearing loss. Interruption of the continuity or mobility of the ossicular chain can also cause CHL; ossicles may be eroded or missing.

##### Symptoms

Patients will report hearing loss, difficulty understanding speech in noisy environments, and occasionally tinnitus. Disorders of the EAC and TM (e.g., cholesteatoma, chronic otitis media) may also cause otorrhea and recurrent infections. Inciting events may be elicited, such as head trauma, recent upper respiratory tract infection, or flight. Causes of potential auditory (eustachian) tube dysfunction should be elicited, such as chronic sinusitis, allergic rhinitis, or cleft palate.

##### Differential Diagnosis

The differential diagnosis of CHL is broad and can involve disorders of the EAC, TM, middle ear, and ossicles. Disorders of the EAC include congenital atresia, cerumen, foreign body, exostosis, osteoma, and neoplasm. Disorders

of the TM include scarring, tympanosclerosis, retraction, and perforation. Disorders of the middle ear can include adhesions and effusions and may stem from underlying auditory tube dysfunction. Disorders that can cause ossicular discontinuity or ossicular erosion are trauma, cholesteatoma, chronic otitis media, and neoplasm. Otosclerosis and malleus fixation can also impair the vibratory potential of the ossicular chain. Third window disorders such as semicircular canal dehiscence can also cause CHL.

## ◆ Evaluation

### Physical Exam

A full head and neck examination should be performed to elicit signs of underlying sinonasal disease, auditory tube dysfunction, or neoplasm. In an adult with persistent unilateral middle ear effusion, nasopharyngoscopy is performed to rule out nasopharyngeal mass causing auditory tube dysfunction. Binocular microscopic evaluation of the EAC and TM is extremely helpful in identifying pathology of CHL, which may otherwise be missed with a handheld otoscopy. Complete removal of cerumen aids in visualization. Pneumatic otoscopy can assist in determining TM mobility and should always be documented. The integrity of the manubrium of the malleus, long process of the incus, and stapes can also be seen on exam. In general, the Weber test with a 512-Hz tuning fork generally lateralizes to the ear with CHL. Depending on the degree of CHL, the Rinne test will demonstrate bone conduction greater than air conduction (“negative Rinne”).

### Imaging

High-resolution computed tomography (CT) of the temporal bones is not necessary in all cases of CHL. CT scan is obtained when there is a chronically draining ear (e.g., chronic otitis media), evidence of cholesteatoma, or mass lesion of the EAC or middle ear or when the cause of CHL is unclear. CT can also be done for surgical planning (e.g., atresia repair). Otosclerotic foci can be demonstrated on CT, although imaging is not necessary in all cases.

### Other Tests

Comprehensive audiometry with both air conduction and bone conduction testing should be performed to determine air–bone gap (ABG), which represents the amount of hearing loss attributed to the conductive mechanism. Tympanometry can be performed to determine the mobility of the drum, detecting perforations and middle ear effusions. Acoustic reflexes should be absent. If acoustic reflexes are present in a patient with a conductive hearing loss, superior semicircular canal dehiscence should be suspected. In this case, cervical vestibular evoked myogenic potential (cVEMP) testing can be helpful, as superior semicircular canal dehiscence syndrome patients typically demonstrate larger amplitudes and lower thresholds on cVEMP testing at 500 Hz.

## ◆ Treatment Options

### Medical

Amplification and observation should be offered to patients as a treatment option. Amplification can include hearing aids and soft band (bone conducting hearing aid). Certain conditions may require surgical intervention prior to amplification (e.g., neoplasm, chronic otitis media, cholesteatoma).

### Surgical

Surgical intervention can potentially improve hearing when there is a significant ABG. The surgery depends on the etiology of the CHL. Osseointegrated implants (see **Chapter 3.5.5**) are an alternative treatment for CHL that should be considered when surgery for the underlying disorder may carry high risk to sensorineural component of hearing or facial nerve (e.g., atresia).

#### *Outer Ear*

- Cerumen removal
- Foreign body removal
- Canalplasty with or without skin graft (for exostosis or stenosis)
- Meatoplasty
- Atresia repair

#### *Tympanic Membrane*

- Myringotomy with or without tympanostomy tube for middle ear effusion
- Paper patch or fat myringoplasty for small perforation (often office procedures)
- Tympanoplasty

#### *Middle Ear*

- Ossicular chain reconstruction using remodeled incus or synthetic prosthesis
- Stapedectomy/stapedotomy
- Exploratory tympanotomy with or without mastoidectomy
  - Lysis of middle ear adhesions
  - Removal of tympanosclerosis
  - Repair of malleus fixation
  - Removal of cholesteatoma
  - Treat chronic otitis media
  - Removal of neoplasm

## ◆ Typical Clinical Pictures

### Otosclerosis

Otosclerosis is a process in which the stapes loses mobility by excessive bony growth at the oval window. It typically causes a CHL with a distinctive dip in sensorineural levels at 2 kHz, known as Carhart notch. Patients may have tinnitus and progressive hearing loss in one or both ears. Otosclerosis has a bimodal age epidemiology; it usually presents in the early twenties or later in the fifties. Early otosclerosis can be managed with fluoride supplementation, but often the CHL will progress and patients will need to consider amplification or stapes surgery. Stapedectomy or stapedotomy involves removing the superstructure of the stapes, and entering into the inner ear, either by making a fenestration in the footplate (stapedotomy), or partially removing it (stapedectomy). A synthetic prosthesis is then placed into the oval window and attached to the incus to reestablish ossicular transmission of sound waves.

### Ossicular Disease

Cholesteatoma, chronic ear infections, and trauma can all cause erosion or disruption of the ossicular chain. After appropriately treating the aforementioned causes, one may evaluate the remaining middle ear structures to determine the appropriate surgical repair. Once the current ossicular status (anatomic and functional integrity of each ossicle) is determined, the appropriate prosthesis can be selected. Prostheses exist for nearly every situation of ossicular disease. Common prostheses include a partial ossicular reconstruction prosthesis (replaces incus and malleus), incus struts (replace incus only), and a total ossicular reconstruction prosthesis (replaces all ossicles).

## ◆ Complications

Obstructive EAC stenosis after canalplasty may require revision canalplasty. Tympanoplasty has a failure rate of ~15%, and revision tympanoplasty may be necessary; factors that may affect success rate is the size of the TM perforation (e.g., subtotal or total), location (e.g., anterior, marginal), and etiology (e.g., chronic otitis media). Revision surgery may be necessary after ossicular chain reconstruction if the prosthesis shifts and no longer conducts sound effectively. Six months is generally considered the earliest time point at which revision surgery should be considered. Progressive hearing loss 10 to 14 days after stapes surgery may indicate reparative granuloma, and middle ear exploration should be considered.

## ◆ Outcome and Follow-Up

Water precautions should be observed in the postoperative period. If the TM is grafted or a stapedotomy is performed, caution regarding airline flight should be considered in the immediate postoperative period to avoid TM movement. A postoperative audiogram to determine level of hearing restoration should be performed 6 to 8 weeks after procedure (adequate time for gelatin packing in the middle ear to absorb).

### 3.5.2 Sensorineural Hearing Loss

#### ◆ Key Features

- Sensorineural hearing loss (SNHL) is caused by the dysfunction of the sensory (cochlea) or neural components of the auditory system.
- Asymmetric SNHL may need to be evaluated by MRI or ABR to rule out retrocochlear pathology.
- Treatment may include amplification, osseointegrated implants, and cochlear implantation

Sensorineural hearing loss (SNHL) can stem from disorders of the cochlea, cochlear nerve, and central auditory processing. It can affect one or both ears. Asymmetrical SNHL is often described as a 10-dB difference in three consecutive pure tone frequencies, a 15-dB difference in two consecutive frequencies, and/or a  $\geq 12\%$  point difference in speech recognition.

#### ◆ Clinical

##### Signs and Symptoms

Patients typically notice difficulty in crowds or with background noise. They may report having to increase the volume of the television or radio to understand. Patients with high-frequency hearing loss may notice difficulty with understanding women or children's voices. Many patients will not realize they have hearing loss until it has progressed significantly, as the hearing loss is usually very gradual. Often, tinnitus is associated with hearing loss and may be the presenting symptom. Other otologic symptoms, such as otorrhea, aural fullness, and vertigo, may also help identify the etiology of the hearing loss. Inciting events or risk factors for hearing may be present, such as head trauma, acoustic trauma, chronic loud noise exposure, family history of hearing loss, ototoxic drug use, prematurity, prenatal/postnatal infection, barotrauma, and autoimmune disease. A full review of systems concentrating on neurologic or rheumatologic symptoms may also imply the cause.

##### Differential Diagnosis for Symmetric SNHL

###### *Presbycusis*

Slowly progressive bilateral SNHL in the older population (presbycusis) is widely prevalent and is the most common form of SNHL. It is often a familial condition. However, a detailed otologic history may find other factors that may contribute to SNHL.

###### *Noise-Induced Hearing Loss*

Exposure to loud noise can lead to permanent hearing threshold shifts. This may happen immediately with extreme exposure (nearby explosion or gunfire) but more commonly occurs slowly over time with repeated exposure to industrial or environmental noise. Patients will often have a typical 4-kHz

**Table 3.9**    Common ototoxic drugs

|                                                                       |
|-----------------------------------------------------------------------|
| Aminoglycoside antibiotics (e.g., streptomycin, gentamicin, neomycin) |
| Platinum-based chemotherapy (especially cisplatin)                    |
| Loop diuretics                                                        |
| Vinca alkaloids (especially vincristine)                              |
| Quinine                                                               |
| Salicylates                                                           |

notch in their audiograms. Patients should be counseled to prevent further noise damage by wearing appropriate hearing protection and minimizing exposure.

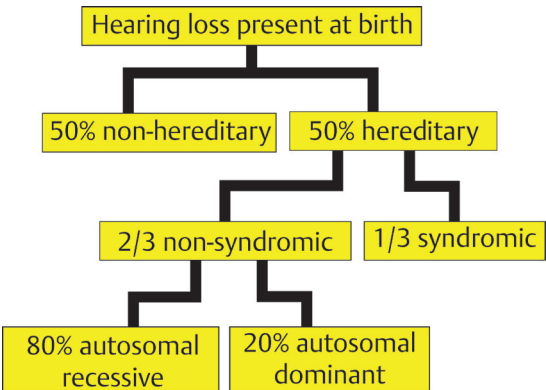
*Ototoxicity*

Exposure to a variety of medications may induce permanent hearing loss (**Table 3.9**). This hearing loss is typically first noted in the highest frequencies and then progresses to lower pitches. Common agents include aminoglycoside antibiotics, vinca alkaloids, and platinum-based chemotherapeutic agents. Careful monitoring of audiograms during therapy allow for early identification of hearing loss. Prolonged use of high-dose loop diuretics (e.g., furosemide) may also lead to hearing loss. Of note, many ototoxic drugs are also nephrotoxic; therefore, renal function studies should be obtained as well.

*Hearing Loss Present at Birth*

See **Fig. 3.11** for a classification overview of hearing loss at birth. About 50% of cases are nonhereditary, from prematurity, sepsis, or TORCH infections (toxoplasmosis, rubella, cytomegalovirus, herpes simplex encephalitis, and otosyphilis). Of the 50% that are congenital hereditary cases, these may be syndromic (one-third of cases) or nonsyndromic (two-thirds of cases); see **Table 3.10**. Of the nonsyndromic cases, ~80% are autosomal recessive.

Hearing loss may be present at birth due to congenital defects in either the structure or the physiology of the inner ear. Cochlear malformations



**Fig. 3.11**    Hearing loss present at birth: a classification overview.

**Table 3.10** Hereditary syndromic hearing loss

| Autosomal dominant | Autosomal recessive   | Sex-linked |
|--------------------|-----------------------|------------|
| Treacher Collins   | Usher                 | Alport     |
| Goldenhar          | Pendred               |            |
| Waardenberg        | Jervell-Lange-Nielsen |            |
| Branchio-oto-renal |                       |            |
| Stickler           |                       |            |

(e.g., Mondini malformation, common cavity, cochlear hypoplasia, cochlear aplasia) may be associated with various syndromes. Many cases of nonsyndromic congenital hearing loss have been attributed to chromosomal defects in the hair cell protein connexin 26 (Cx 26). Most congenital cases are now discovered early due to universal newborn screening programs. However, certain types, such as enlarged vestibular aqueduct syndrome, may not manifest until later in life. Auditory neuropathy can also cause unilateral or bilateral SNHL.

*Metabolic*

Symmetric bilateral rapidly progressive hearing loss may be caused by a variety of systemic diseases, including autoimmune disease, thyroid disease, or vitamin B<sub>12</sub> or folate deficiencies.

*Other*

Cochlear otosclerosis, primary autoimmune inner ear disease, barotrauma, perilymph fistula, migraine, vertebrobasilar occlusion, Ménière's disease, labyrinthitis ossificans from meningitis, and a variety of autoimmune disorders may all be associated with SNHL.

**Differential Diagnosis for Asymmetric Hearing Loss**

*Tumors*

Neoplasms of the cerebellopontine angle and/or internal auditory canal, such as vestibular schwannomas and meningiomas, may cause unilateral SNHL. Endolymphatic sac tumors, especially in patients with a history suggestive of Von Hippel–Lindau's syndrome, can also cause unilateral SNHL. Disorders of the otic capsule bone (e.g. fibrous dysplasia, Paget's disease, osteogenesis imperfecta) may also cause a sensory component to an asymmetric hearing loss that more commonly is associated with conductive hearing loss. Metastasis to the temporal bone should be considered in patients with a history of malignancy such as breast, prostate, thyroid, lung, and renal cancer.

*Ménière's Disease*

Active Ménière's disease typically causes a fluctuating low-frequency hearing loss. As the disease progresses, higher frequencies are affected and can progress to severe levels. Bilateral Ménière can also occur, but onset of symptoms in the second ear is generally years after the first.



### *Infectious*

Meningitis may lead to the spread of bacteria to the inner ear through the cochlear aqueduct, which can lead to cochlear fibrosis and ossification. Congenital TORCH infections can cause early-onset SNHL. Viral labyrinthitis can also cause SNHL. Infectious causes may lead to symmetric or asymmetric losses.

### *Traumatic*

Fractures of the temporal bone involving the otic capsule usually lead to profound hearing loss. Leakage of perilymph from the oval or round windows can cause progressive hearing loss and dizziness. Strong Valsalva during heavy lifting, head trauma, or barotrauma may initiate these perilymphatic fistulas.

### *Neurologic Disease*

Multiple sclerosis is well known for causing a myriad of neurologic symptoms, including hearing loss. Cerebrovascular disease leading to brainstem stroke or impaired blood supply to the cochlea can also cause hearing loss.

## ◆ Evaluation

### **Physical Exam**

A thorough head and neck examination should be performed. The ear should be examined for any potential cause of CHL, such as cerumen, tympanic perforation, middle ear effusion, and ossicular chain disorder. Tuning fork tests may be done to document laterality and nature of hearing loss. In general, the Weber test using a 512-Hz tuning fork will lateralize toward the better hearing ear (in the absence of conductive hearing loss). Rinne test is positive (air conduction > bone conduction).

### **Imaging**

Temporal bone computed tomography (CT) is helpful in identifying inner ear abnormalities associated with hearing loss in children (e.g., enlarged vestibular aqueduct, inner ear malformation). Both magnetic resonance imaging (MRI) and CT can identify labyrinthitis ossificans. CT is helpful in identifying temporal fracture as an etiology for disorders of the otic capsule bone. To assess for retrocochlear pathology in asymmetric hearing loss, MRI scanning of the brain with fine cuts through the internal auditory canals can be performed with and without contrast media. Heavily weighted T2 fast spin echo series has also been used to screen for retrocochlear pathology.

### **Labs**

There is controversy regarding laboratory testing. For pediatric hearing loss, consider urinalysis (for Alport and branchio-oto-renal syndrome), electrocardiogram (ECG, for Jervell-Lange-Nielsen's syndrome), and referral for ophthalmology exam (to rule out retinitis pigmentosa). Additional testing can include rapid plasma reagin (RPR) or fluorescent treponemal antibody absorption (FTA-ABS), toxoplasmosis, cytomegalovirus-related tests, thyroid function testing, erythrocyte sedimentation rate (ESR), antinuclear antibody testing, rheumatoid factor, liver function tests, and complete blood count (CBC).

In adults with bilateral progressive or fluctuating hearing loss, consider antinuclear antibody, rheumatoid factor, anti-neutrophil cytoplasmic antibody (ANCA), ESR, and syphilis testing (RPR or FTA-ABS). Lyme serology can be ordered if suspected as well. The utility of a 68-kD antibody testing for diagnosing autoimmune inner ear disease is controversial.

## Other Tests

Pure tone audiometry is the standard for documentation of hearing loss. Speech reception thresholds and word discrimination testing should also be performed. Patients may also be screened with otoacoustic emission testing and auditory brainstem response (ABR) testing when audiometry cannot be performed. ABR can also be used as an alternative to MRI for evaluating asymmetric hearing loss. Referral for genetic testing can be considered. Hearing loss can be classified by its severity:

- Mild (15–40 dB)
- Moderate (41–55 dB)
- Moderate to severe (56–70 dB)
- Severe (70–90 dB)
- Profound (> 90 dB)

## ◆ Treatment Options

### Medical

Bilateral symmetric SNHL with >50% speech discrimination should be offered hearing aid amplification. Depending on the pattern of hearing loss, middle ear implants or implantable hearing aids are alternative therapies. When patients have profound SNHL or word discrimination is very poor, hearing aid trial and cochlear implant evaluation should be considered.

Patients with asymmetric SNHL can benefit from standard hearing aids. Contralateral routing of sound (CROS), Bi-CROS (if the good ear requires amplification), and osseointegrated implants are treatment options for single-sided deafness.

Medical treatment of the underlying etiology (when known) may be important in controlling disease progression. Treatment of sudden hearing loss can include oral and/or intratympanic steroid administrations. Treatment of associated tinnitus is discussed in **Chapter 3.7**.

### Surgical

Patients with profound bilateral loss that is not helped by hearing aids may be candidates for cochlear implantation (see **Chapter 3.5.4**). Patients with profound unilateral loss can be candidates for osseointegrated bone-anchored hearing devices (see **Chapter 3.5.5**).

## ◆ Outcome and Follow-Up

Frequent adjustments and cleaning of hearing aids will ensure better patient compliance and satisfaction. Yearly audiograms should be performed on patients with identified SNHL to monitor for progression.